SECTION 1

Character RNNs and text generation

- **Char-RNN:** A model trained to predict the next character can compose novel text; the chapter walks from tokenizing Shakespeare to building `tf.data` sliding windows with non-overlapping splits for train/validation/test to avoid leakage in sequential data.
- **Model and sampling:** Stacked GRU layers with a time-distributed softmax predict the next symbol; at generation time temperature-scaled categorical sampling trades repetition for diversity instead of always picking the argmax.
- **Stateful RNNs:** Carrying hidden state across batches (with aligned, non-shuffled windows) helps learn patterns longer than the truncated BPTT window, at the cost of awkward batching and inference constraints unless weights are copied to a stateless twin.

SECTION 2

Sentiment analysis and variable-length text

- **IMDb pipeline:** Word-level RNNs on movie reviews use embeddings, stacked GRUs, and binary classification; the chapter contrasts Keras's pre-indexed IMDb set with TensorFlow Datasets, vocabulary tables, and out-of-vocabulary buckets.
- **Masking:** `mask_zero=True` on `Embedding` (or explicit masks) tells downstream layers to ignore padding so the model does not waste capacity on filler tokens.
- **Transfer:** TensorFlow Hub sentence encoders supply pretrained word statistics, so a small classifier on top can train quickly with little labeled data.

SECTION 3

Encoder–decoder neural machine translation

- **Seq2seq:** An encoder compresses a source sentence; a decoder emits the translation one target word at a time, fed shifted targets while training and its own previous outputs at inference.
- **Practical constraints:** Length bucketing, masking past EOS in the loss, sampled softmax for huge vocabularies, bidirectional encoders, and beam search at decode time improve quality without retraining.
- **Tooling:** TensorFlow Addons supplies training samplers (including scheduled sampling), decoders, and beam-search wrappers for production-style NMT.

SECTION 4

Attention mechanisms

- **Motivation:** Long encoder–decoder paths blur information; attention lets each decoder step re-weight all encoder outputs, shortening the effective path for aligning source and target tokens (e.g., "milk" → "lait").
- **Variants:** Bahdanau (concatenative) and Luong (dot-product) attention differ in how alignment scores are computed; attention also supports visual captioning and explainability by showing what the model focused on.
- **Beyond translation:** The same idea applies wherever a model must pick relevant context from a long input.

SECTION 5

The Transformer architecture

- **Attention-only stack:** The 2017 Transformer removes recurrence, using encoder and decoder stacks of multi-head self-attention, feed-forward blocks, residuals, and layer normalization.
- **Positional information:** Sine/cosine positional encodings (or learned positions) inject order because self-attention is permutation-invariant without them.
- **Multi-head attention:** Parallel scaled dot-product heads attend in different representation subspaces; masked decoder attention prevents peeking at future tokens during training.

SECTION 6

Large language models (2018–2019)

- **Contextual embeddings and pretraining:** ELMo supplies context-dependent vectors; ULMFiT and GPT show unsupervised pretrain + fine-tune (and GPT-2 zero-shot use) on Transformer-like stacks.
- **BERT:** Masked language modeling and next-sentence prediction yield bidirectional representations that fine-tune strongly on classification, entailment, and question answering.
- **Trend:** Subword tokenization plus self-supervised Transformers became the default NLP recipe, though the chapter notes architecture fashion can shift.